# OV2640 Color CMOS UXGA (2.0 MegaPixel) CameraChip™ with OmniPixel2™ Technology

## General Description

The OV2640 CameraChip™ is a low voltage CMOS image sensor that provides the full functionality of a single-chip UXGA (1632x1232) camera and image processor in a small footprint package. The OV2640 provides full-frame, sub-sampled, scaled or windowed 8-bit/10-bit images in a wide range of formats, controlled through the Serial Camera Control Bus (SCCB) interface.

This product has an image array capable of operating at up to 15 frames per second (fps) in UXGA resolution with complete user control over image quality, formatting and output data transfer. All required image processing functions, including exposure control, gamma, white balance, color saturation, hue control, white pixel canceling, noise canceling, and more, are also programmable through the SCCB interface. The OV2640 also includes a compression engine for increased processing power. In addition, OmniVision CameraChips use proprietary sensor technology to improve image quality by reducing or eliminating common lighting/electrical sources of image contamination, such as fixed pattern noise, smearing, etc., to produce a clean, fully stable color image.

**Note:** The OV2640 uses a lead-free package.

## Features

- High sensitivity for low-light operation
- Low operating voltage for embedded portable apps
- Standard SCCB interface
- Output support for Raw RGB, RGB (RGB565/555), GRB422, YUV (422/420) and YCbCr (4:2:2) formats
- Supports image sizes: UXGA, SXGA, SVGA, and any size scaling down from SXGA to 40x30
- VarioPixel® method for sub-sampling
- Automatic image control functions including Automatic Exposure Control (AEC), Automatic Gain Control (AGC), Automatic White Balance (AWB), Automatic Band Filter (ABF), and Automatic Black-Level Calibration (ABLC)
- Image quality controls including color saturation, gamma, sharpness (edge enhancement), lens correction, white pixel canceling, noise canceling, and 50/60 Hz luminance detection
- Line optical black level output capability
- Video or snapshot operation
- Zooming, panning, and windowing functions
- Internal/external frame synchronization
- Variable frame rate control
- Supports LED and flash strobe mode
- Supports scaling
- Supports compression
- Embedded microcontroller

## Ordering Information

| Product | Package |
|---|---|
| OV02640-VL9A (Color, lead-free) | 38-pin CSP2 |

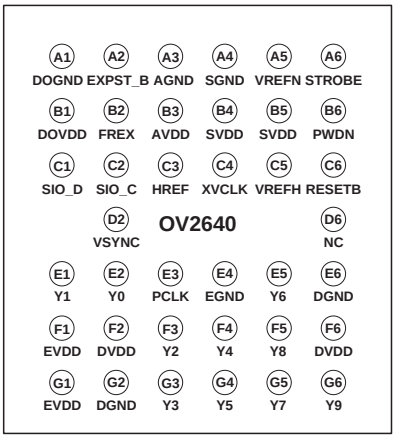
## Applications

- Cellular and Camera Phones
- Toys
- PC Multimedia
- Digital Still Cameras

## Key Specifications

| | | |
|---|---|---|
| **Array Size** | **UXGA** | 1600 x 1200 |
| **Power Supply** | **Core** | 1.3VDC ± 5% |
| | **Analog** | 2.5 ~ 3.0VDC |
| | **I/O** | 1.7V to 3.3V |
| **Power Requirements** | **Active** | 125 mW (for 15 fps, UXGA YUV mode) |
| | | 140 mW (for 15 fps, UXGA compressed mode) |
| | **Standby** | 600 µA |
| **Temperature Range** | **Stable Image** | 0°C to 50°C |
| **Output Formats (8-bit)** | | • YUV(422/420)/YCbCr422<br>• RGB565/555<br>• 8-bit compressed data<br>• 8-/10-bit Raw RGB data |
| **Lens Size** | | 1/4" |
| **Chief Ray Angle** | | 25° non-linear |
| **Maximum Image Transfer Rate** | **UXGA/SXGA** | 15 fps |
| | **SVGA** | 30 fps |
| | **CIF** | 60 fps |
| **Sensitivity** | | 0.6 V/Lux-sec |
| **S/N Ratio** | | 40 dB |
| **Dynamic Range** | | 50 dB |
| **Scan Mode** | | Progressive |
| **Maximum Exposure Interval** | | 1247 x $t_{ROW}$ |
| **Gamma Correction** | | Programmable |
| **Pixel Size** | | 2.2 µm x 2.2 µm |
| **Dark Current** | | 15 mV/s at 60°C |
| **Well Capacity** | | 12 Ke |
| **Fixed Pattern Noise** | | <1% of $V_{PEAK-TO-PEAK}$ |
| **Image Area** | | 3590 µm x 2684 µm |
| **Package Dimensions** | | 5725 µm x 6285 µm |

**Figure 1 OV2640 Pin Diagram (Top View)**



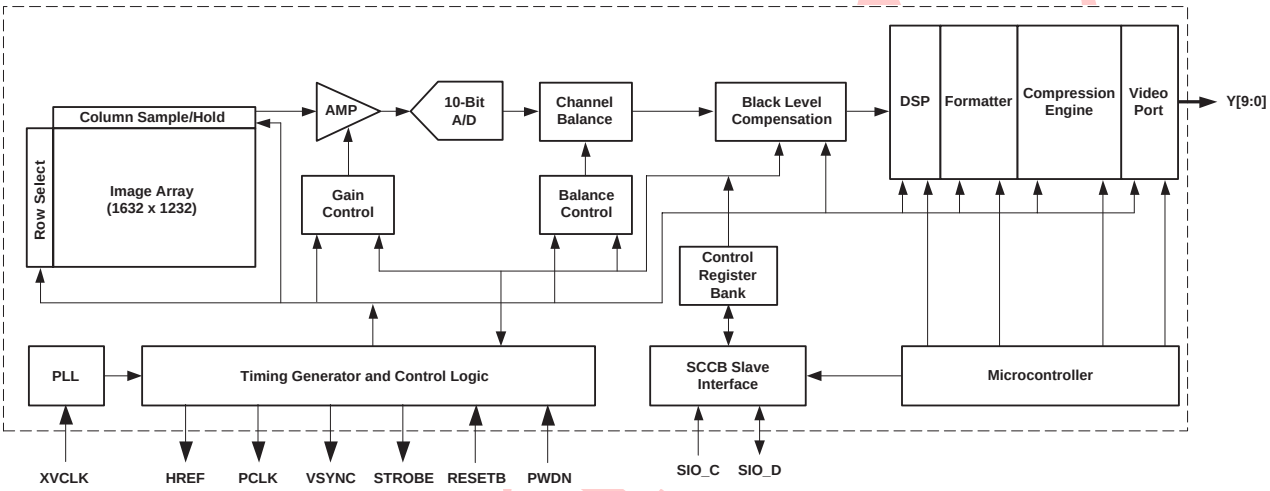

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## Functional Description

Figure 2 shows the functional block diagram of the OV2640 image sensor. The OV2640 includes:

- Image Sensor Array (1632 x 1232 total image array)
- Analog Signal Processor
- 10-Bit A/D Converters
- Digital Signal Processor (DSP)
- Output Formatter
- Compression Engine
- Microcontroller
- SCCB Interface
- Digital Video Port

**Figure 2 Functional Block Diagram**

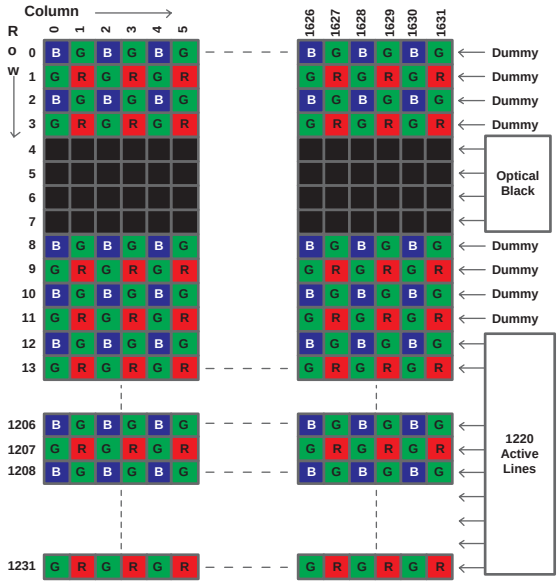



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## Image Sensor Array

The OV2640 sensor has an image array of 1632 columns by 1232 rows (2,010,624 pixels). Figure 3 shows a cross-section of the image sensor array.

**Figure 3 Sensor Array Region Color Filter Layout**




The color filters are arranged in a Bayer pattern. The primary color BG/GR array is arranged in line-alternating fashion. Of the 2,010,624 pixels, 1,991,040 (1632x1220) are active. The other pixels are used for black level calibration and interpolation.

The sensor array design is based on a field integration read-out system with line-by-line transfer and an electronic shutter with a synchronous pixel read-out scheme.

## Analog Amplifier

When the column sample/hold circuit has sampled one row of pixels, the pixel data will shift out one-by-one into an analog amplifier.

### Gain Control

The amplifier gain can either be programmed by the user or controlled by the internal automatic gain control circuit (AGC).

## 10-Bit A/D Converters

After the analog amplifier, the bayer pattern Raw signal is fed to two 10-bit analog-to-digital (A/D) converters, one for G channel and one shared by the BR channels. These A/D converters operate at speeds up to 20 MHz and are fully synchronous to the pixel rate (actual conversion rate is related to the frame rate).

## Channel Balance

The amplified signals are then balanced with a channel balance block. In this block, the Red/Blue channel gain is increased or decreased to match Green channel luminance level.

### Balance Control

Channel Balance can be done manually by the user or by the internal automatic white balance (AWB) controller.

## Black Level Compensation

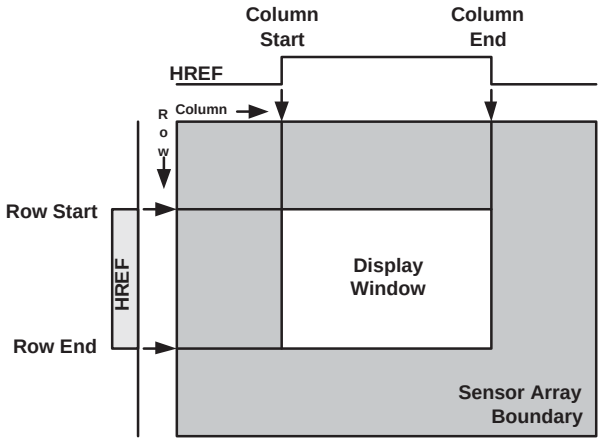
After the pixel data has been digitized, black level calibration can be applied before the data is output. The black level calibration block subtracts the average signal level of optical black pixels to compensate for the dark current in the pixel output. The user can disable black level calibration.

## Windowing

The OV2640 allows the user to define window size or region of interest (ROI), as required by the application. Window size setting (in pixels) ranges from 2 x 4 to 1632 x 1220 (UXGA) or 2 x 2 to 818 x 610 (SVGA), and 408 x 304 (CIF), and can be anywhere inside the 1632 x 1220 boundary. Note that modifying window size or window position does not alter the frame or pixel rate. The windowing control merely alters the assertion of the HREF signal to be consistent with the programmed horizontal and vertical ROI. The default window size is 1600 x 1200. Refer to Figure 4 and registers HREFST, HREFEND, REG32, VSTRT, VEND, and COM1 for details.

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**Figure 4 Windowing**




## Zooming and Panning Mode

The OV2640 provides zooming and panning modes. The user can select this mode under SVGA/CIF mode timing. The related zoom ratios will be 2:1 of UXGA for SVGA and 4:1 of UXGA for CIF. Registers ZOOMS[7:0] (0x49) and COM19[1:0] (0x48) define the vertical line start point. Register ARCOM2[2] (0x34) defines the horizontal start point.

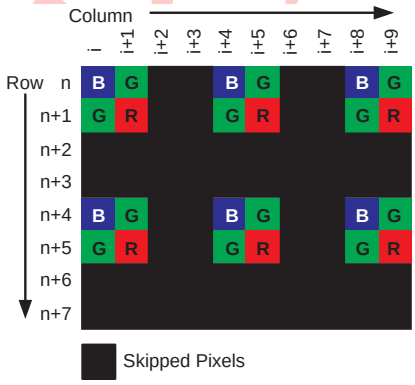
## Sub-sampling Mode

The OV2640 supports two sub-sampling modes. Each sub-sampling mode has different resolution and maximum frame rate. These modes are described in the following sections.

### *SVGA mode*

The OV2640 can be programmed to output 800 x 600 (SVGA) sized images for applications where higher resolution image capture is not required. In this mode, both horizontal and vertical pixels will be sub-sampled with an aspect ratio of 4:2 as shown in Figure 5.

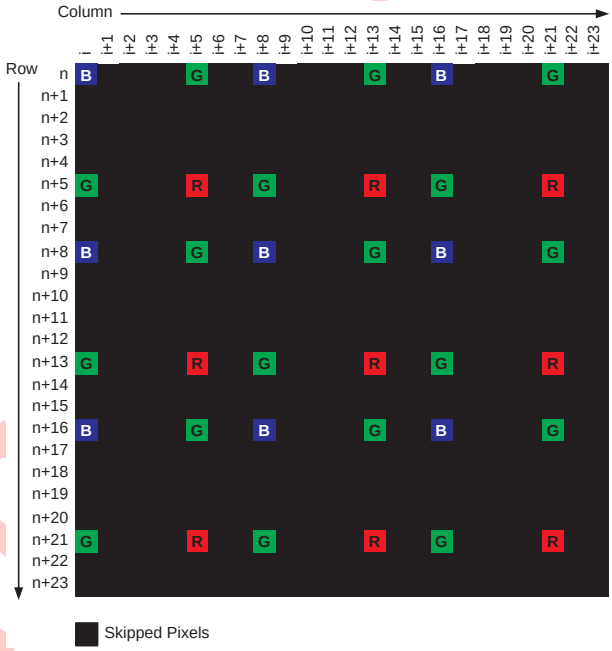
**Figure 5 SVGA Sub-Sampling Mode**




### *CIF Mode*

The OV2640 can also operate at a higher frame rate to output 400 x 296 sized images. Figure 6 shows the sub-sampling diagram in both horizontal and vertical directions for CIF mode.

**Figure 6 CIF Sub-Sampling Mode**




## Timing Generator and Control Logic

In general, the timing generator controls the following:

- Frame Exposure Mode Timing
- Frame Rate Adjust
- Frame Rate Timing

## Frame Exposure Mode Timing

The OV2640 supports frame exposure mode. Typically, the frame exposure mode must work with the aid of an external shutter.

The frame exposure pin, FREX (pin B2), is the frame exposure mode enable pin and the EXPST_B pin (pin A2) serves as the sensor's exposure start trigger. When the external master device asserts the FREX pin high, the sensor array is quickly pre-charged and stays in reset mode until the EXPST_B pin goes low (sensor exposure time can be defined as the period between EXPST_B low and shutter close). After the FREX pin is pulled low, the video data stream is then clocked to the output port in a line-by-line manner. After completing one frame of data

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output, the OV2640 will output continuous live video data unless in single frame transfer mode. Figure 18 and Figure 19 show the detailed timing and Table 11 shows the timing specifications for this mode.

## Frame Rate Adjust

The OV2640 offers three methods for frame rate adjustment:

- Clock prescaler: (see "CLKRC" on page 23)
  By changing the system clock divide ratio and PLL, the frame rate and pixel rate will change together. This method can be used for dividing the frame/pixel rate by: 1/2, 1/3, 1/4 … 1/64 of the input clock rate.
- Line adjustment: (see "REG2A" on page 26 and "FRARL" on page 26)
  By adding a dummy pixel timing in each line (between HSYNC and pixel data out), the frame rate can be changed while leaving the pixel rate as is.
- Vertical sync adjustment:
  By adding dummy line periods to the vertical sync period (see "ADDVSL" on page 26 and "ADDVSH" on page 26 or see "FLL" on page 27 and "FLH" on page 27), the frame rate can be altered while the pixel rate remains the same.

## Frame Rate Timing

Default frame timing is illustrated in Figure 15, Figure 16, and Figure 17. Refer to Table 1 for the actual pixel rate at different frame rates.

**Table 1 Frame/Pixel Rates in UXGA Mode**

| Frame Rate (fps) | 15 | 7.5 | 2.5 | 1.25 |
|---|---|---|---|---|
| PCLK (MHz) | 36 | 18 | 6 | 3 |

## Digital Signal Processor (DSP)

This block controls the interpolation from Raw data to RGB and some image quality control.

- Edge enhancement (a two-dimensional high pass filter)
- Color space converter (can change Raw data to RGB or YUV/YCbCr)
- RGB matrix to eliminate color cross talk
- Hue and saturation control
- Programmable gamma control
- Transfer 10-bit data to 8-bit
- White pixel canceling
- De-noise

## Output Formatter

This block controls all output and data formatting required prior to sending the image out.

## Scaling Image Output

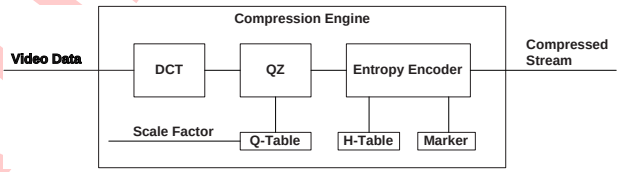
The OV2640 is capable of scaling down the image size from CIF to 40x30. By using SCCB registers, the user can output the desired image size. At certain image sizes, HREF is not consistent in a frame.

## Compression Engine

As shown in Figure 7, the Compression Engine consists of three major blocks:

- DCT
- QZ
- Entropy Encoder

**Figure 7 Compression Engine Block Diagram**




## Microcontroller

The OV2640 embeds an 8-bit microcontroller with 512-byte data memory and 4 KB program memory. It provides the flexibility of decoding protocol commands from the host for controlling the system, as well as the ability to fine tune image quality.

## SCCB Interface

The Serial Camera Control Bus (SCCB) interface controls the CAMERACHIP operation. Refer to *OmniVision Technologies Serial Camera Control Bus (SCCB) Specification* for detailed usage of the serial control port.

## Slave Operation Mode

The OV2640 can be programmed to operate in slave mode (default is master mode).

When used as a slave device, COM7[3] (0x12), CLKRC[6] (0x11), and COM2[2] (0x09) register bits should be set to

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"1" and the OV2640 will use PWDN and RESETB pins as vertical and horizontal synchronization triggers supplied by a master device. The master device must provide the following signals:

1. System clock MCLK to XVCLK pin
2. Horizontal sync MHSYNC to RESETB pin
3. Vertical frame sync MVSYNC to PWDN pin

See Figure 8 for slave mode connections and Figure 9 for detailed timing considerations.

**Figure 8 Slave Mode Connection**

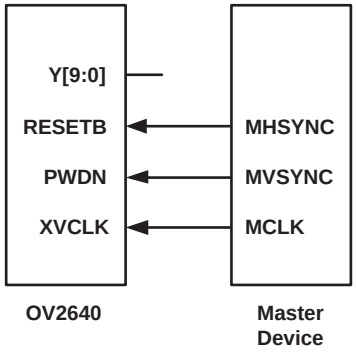
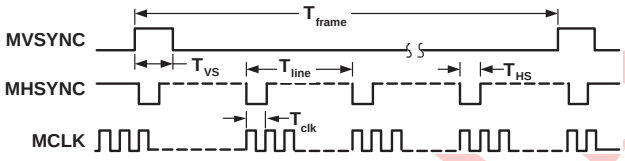



**Figure 9 Slave Mode Timing**




**NOTE:**

1) $T_{HS} > 6\ T_{clk}$, $Tvs > T_{line}$

2) $T_{line} = 1922 \times T_{clk}$ (UXGA); $T_{line} = 1190 \times T_{clk}$ (SVGA); $T_{line} = 595 \times T_{clk}$ (CIF)

3) $T_{frame} = 1248 \times T_{line}$ (UXGA); $T_{frame} = 672 \times T_{line}$ (SVGA); $T_{frame} = 336 \times T_{line}$ (CIF)

## Strobe Mode

The OV2640 has a Strobe mode that allows it to work with an external flash and LED.

## Reset

The OV2640 includes a RESETB pin (pin C6) that forces a complete hardware reset when it is pulled low (GND). The OV2640 clears all registers and resets them to their default values when a hardware reset occurs. A reset can also be initiated through the SCCB interface.

## Power Down Mode

Two methods are available to place the OV2640 into power-down mode: hardware power-down and SCCB software power-down.

To initiate hardware power-down, the PWDN pin (pin B6) must be tied to high. When this occurs, the OV2640 internal device clock is halted and all internal counters are reset. The current draw is less than 15 µA in this standby mode.

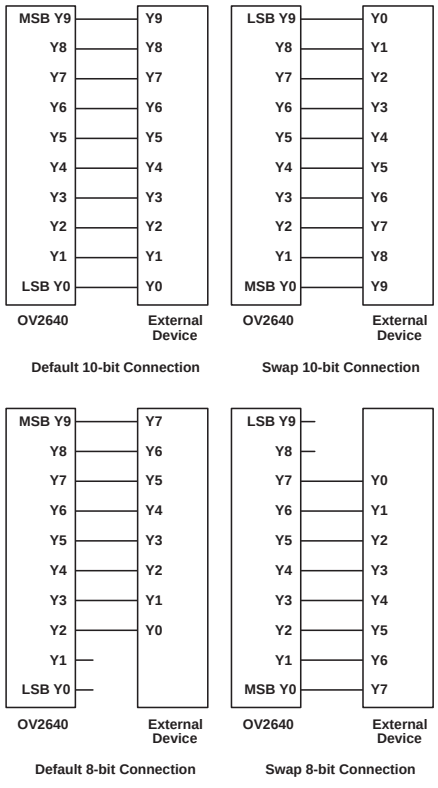
Executing a software power-down through the SCCB interface suspends internal circuit activity but does not halt the device clock. The current requirements drop to less than 1 mA in this mode. All register content is maintained in standby mode.

## Digital Video Port

### MSB/LSB Swap

The OV2640 has a 10-bit digital video port. The MSB and LSB can be swapped with the control registers. Figure 10 shows some examples of connections with external devices.

**Figure 10 Connection Examples**




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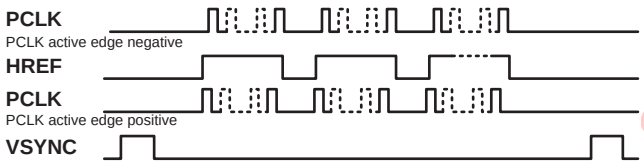
## Line/Pixel Timing

The OV2640 digital video port can be programmed to work in either master or slave mode.

In both master and slave modes, pixel data output is synchronous with PCLK (or MCLK if port is a slave), HREF, and VSYNC. The default PCLK edge for valid data is the negative edge but may be programmed using register COM10[4] for the positive edge. Basic line/pixel output timing and pixel timing specifications are shown in Figure 14 and Table 10.

Also, using register COM10[5], PCLK output can be gated by the active video period defined by the HREF signal. See Figure 11 for details.

**Figure 11 PCLK Output Only at Valid Pixels**




The specifications shown in Table 10 apply for DVDD = +1.2 V, DOVDD = +2.8 V, $T_A$ = 25°C, sensor working at 15 fps, external loading = 20 pF.

### *Pixel Output Pattern*

Table 2 shows the output data order from the OV2640. The data output sequence following the first HREF and after VSYNC is: $B_{0,0}$ $G_{0,1}$ $B_{0,2}$ $G_{0,3}$... $B_{0,1598}$ $G_{0,1599}$. After the second HREF the output is $G_{1,0}$ $R_{1,1}$ $G_{1,2}$ $R_{1,3}$... $G_{1,1598}$ $R_{1,1599}$..., etc. If the OV2640 is programmed to output SVGA resolution data, horizontal and vertical sub-sampling will occur. The default output sequence for the first line of output will be: $B_{0,0}$ $G_{0,1}$ $B_{0,4}$ $G_{0,5}$... $B_{0,1596}$ $G_{0,1597}$. The second line of output will be: $G_{1,0}$ $R_{1,1}$ $G_{1,4}$ $R_{1,5}$... $G_{1,1596}$ $R_{1,1597}$.

**Table 2 Data Pattern**

| R/C | 0 | 1 | 2 | 3 | . . . | 1598 | 1599 |
|---|---|---|---|---|---|---|---|
| **0** | $B_{0,0}$ | $G_{0,1}$ | $B_{0,2}$ | $G_{0,3}$ | . . . | $B_{0,1598}$ | $G_{0,1599}$ |
| **1** | $G_{1,0}$ | $R_{1,1}$ | $G_{1,2}$ | $R_{1,3}$ | . . . | $G_{1,1598}$ | $R_{1,1599}$ |
| **2** | $B_{2,0}$ | $G_{2,1}$ | $B_{2,2}$ | $G_{2,3}$ | . . . | $B_{2,1598}$ | $G_{2,1599}$ |
| **3** | $G_{3,0}$ | $R_{3,1}$ | $G_{3,2}$ | $R_{3,3}$ | . . . | $G_{3,1598}$ | $R_{3,1599}$ |
| **.** | | | | | . | | |
| **.** | | | | | . | | |
| **1198** | $B_{1198,0}$ | $G_{1198,1}$ | $B_{1198,2}$ | $G_{1198,3}$ | . . . | $B_{1198,1598}$ | $G_{1198,1599}$ |
| **1199** | $G_{1199,0}$ | $R_{1199,1}$ | $G_{1199,2}$ | $R_{1199,3}$ | . . . | $G_{1199,1598}$ | $R_{1199,1599}$ |

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## Pin Description

**Table 3 Pin Description**

| Pin Location | Name | Pin Type | Function/Description |
|---|---|---|---|
| A1 | DOGND | Ground | Ground for digital video port |
| A2 | EXPST_B | Input | Snapshot Exposure Start Trigger<br>0: Sensor starts exposure (only effective in snapshot mode)<br>1: Sensor stays in reset mode<br>Note: There is no internal pull-up/pull-down resistor. |
| A3 | AGND | Ground | Ground for analog circuit |
| A4 | SGND | Ground | Ground for sensor array |
| A5 | VREFN | Reference | Internal analog reference - connect to ground using a 0.1 µF capacitor |
| A6 | STROBE | I/O | Flash control output<br>Default: Input<br>Note: There is no internal pull-up/pull-down resistor. |
| B1 | DOVDD | Power | Power for digital video port |
| B2 | FREX | Input | Snapshot trigger - use to activate a snapshot sequence<br>Note: There is no internal pull-up/pull-down resistor. |
| B3 | AVDD | Power | Power for analog circuit |
| B4 | SVDD | Power | Power for sensor array |
| B5 | SVDD | Power | Power for sensor array |
| B6 | PWDN | Input | Power-down mode enable, active high<br>Note: There is an internal pull-down resistor. |
| C1 | SIO_D | I/O | SCCB serial interface data I/O |
| C2 | SIO_C | Input | SCCB serial interface clock input<br>Note: There is no internal pull-up/pull-down resistor. |
| C3 | HREF | I/O | Horizontal reference output<br>Default: Input<br>Note: There is no internal pull-up/pull-down resistor. |
| C4 | XVCLK | Input | System clock input<br>Note: There is no internal pull-up/pull-down resistor. |
| C5 | VREFH | Reference | Internal analog reference - connect to ground using a 0.1 µF capacitor |
| C6 | RESETB | Input | Reset mode, active low<br>Note: There is an internal pull-up resistor. |
| D2 | VSYNC | I/O | Vertical synchronization output<br>Default: Input<br>Note: There is no internal pull-up/pull-down resistor. |
| D6 | NC | – | No connection |
| E1 | Y1 | I/O | Video port output bit[1]<br>Default: Input<br>Note: There is no internal pull-up/pull-down resistor. |

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**Table 3 Pin Description**

| Pin Location | Name | Pin Type | Function/Description |
|---|---|---|---|
| E2 | Y0 | I/O | Video port output bit[0]<br>Default: Input<br>Note: There is no internal pull-up/pull-down resistor. |
| E3 | PCLK | I/O | Pixel clock output<br>Default: Input<br>Note: There is no internal pull-up/pull-down resistor. |
| E4 | EGND | Ground | Ground for internal regulator |
| E5 | Y6 | I/O | Video port output bit[6]<br>Default: Input<br>Note: There is no internal pull-up/pull-down resistor. |
| E6 | DGND | Ground | Ground for digital core |
| F1 | EVDD | Power | Power for internal regulator |
| F2 | DVDD | Power | Sensor digital power (Core) |
| F3 | Y2 | I/O | Video port output bit[2]<br>Default: Input<br>Note: There is no internal pull-up/pull-down resistor. |
| F4 | Y4 | I/O | Video port output bit[4]<br>Default: Input<br>Note: There is no internal pull-up/pull-down resistor. |
| F5 | Y8 | I/O | Video port output bit[8]<br>Default: Input<br>Note: There is no internal pull-up/pull-down resistor. |
| F6 | DVDD | Power | Sensor digital power (Core) |
| G1 | EVDD | Power | Power for internal regulator |
| G2 | DGND | Ground | Ground for digital core |
| G3 | Y3 | I/O | Video port output bit[3]<br>Default: Input<br>Note: There is no internal pull-up/pull-down resistor. |
| G4 | Y5 | I/O | Video port output bit[5]<br>Default: Input<br>Note: There is no internal pull-up/pull-down resistor. |
| G5 | Y7 | I/O | Video port output bit[7]<br>Default: Input<br>Note: There is no internal pull-up/pull-down resistor. |
| G6 | Y9 | I/O | Video port output bit[9]<br>Default: Input<br>Note: There is no internal pull-up/pull-down resistor. |

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**Figure 12 Pinout Diagram**

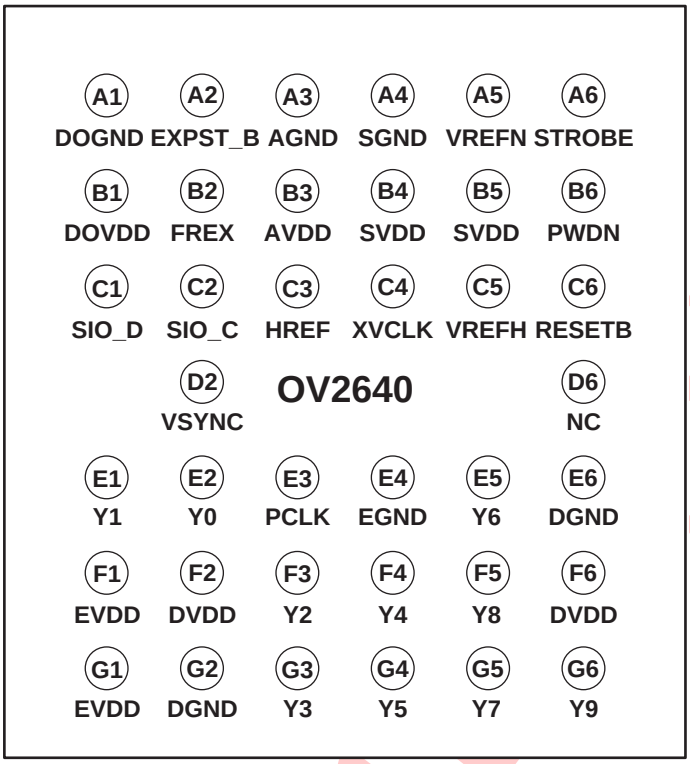



**Table 4 Ball Matrix**

| | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **A** | DOGND | EXPST_B | AGND | SGND | VREFN | STROBE |
| **B** | DOVDD | FREX | AVDD | SVDD | SVDD | PWDN |
| **C** | SIO_D | SIO_C | HREF | XVCLK | VREFN | RESETB |
| **D** | | VSYNC | | | | NC |
| **E** | Y1 | Y0 | PCLK | EGND | Y6 | DGND |
| **F** | EVDD | DVDD | Y2 | Y4 | Y8 | DVDD |
| **G** | EVDD | DGND | Y3 | 75 | Y7 | Y9 |

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# Electrical Characteristics

**Table 5 Absolute Maximum Ratings**

| | | |
|---|---|---|
| **Ambient Storage Temperature** | | -40ºC to +95ºC |
| **Supply Voltages (with respect to Ground)** | **$V_{DD-A}$** | 4.5V |
| | **$V_{DD-C}$** | 3V |
| | **$V_{DD-IO}$** | 4.5V |
| **All Input/Output Voltages (with respect to Ground)** | | -0.3V to $V_{DD-IO}$+1V |
| **Lead-free Temperature, Surface-mount process** | | 245ºC |

**NOTE:** Exceeding the Absolute Maximum ratings shown above invalidates all AC and DC electrical specifications and may result in permanent device damage.

**Table 6 DC Characteristics (-30°C < $T_A$ < 70°C)**

| Symbol | Parameter | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| **Supply** | | | | | |
| $V_{DD-A}$ | Supply voltage | 2.5[a] | 2.8 | 3.0 | V |
| $V_{DD-D}$ | Supply voltage | 1.24 | 1.3 | 1.36 | V |
| $V_{DD-IO}$ | Supply voltage[b] | 1.71 | 2.8 | 3.3 | V |
| $I_{DDA-A}$ | Active (Operating) Current[c] | | 30 | 40 | mA |
| $I_{DDA-D}$ | Active (Operating) Current[c] | | 30 (YUV)<br>45 (Compressed) | 40 (YUV)<br>60 (Compressed) | mA |
| $I_{DDA-IO}$ | Active (Operating) Current[c] | | 6 | 15 | mA |
| $I_{DDS-SCCB}$ | Standby Current[d] | | 1 | 2 | mA |
| $I_{DDS-PWDN}$ | | | 600 | 1200 | µA |
| **Digital Inputs** | | | | | |
| $V_{IL}$ | Input voltage LOW | | | 0.54 | V |
| $V_{IH}$ | Input voltage HIGH | 1.26 | | | V |
| $C_{IN}$ | Input capacitor | | | 10 | pF |
| **Digital Outputs (standard loading 25 pF)** | | | | | |
| $V_{OH}$ | Output voltage HIGH | 1.62 | | | V |
| $V_{OL}$ | Output voltage LOW | | | 0.18 | V |
| **Serial Interface Inputs** | | | | | |
| $V_{IL}$ | SIO_C and SIO_D | -0.5 | 0 | 0.54 | V |
| $V_{IH}$ | SIO_C and SIO_D | 1.26 | 1.8 | 2.3 | V |

a. If using internal regulator for DVDD, $V_{DD-A}$ requires greater than or equal to 2.65V

b. 1.8V I/O is supported. Contact your local OmniVision FAE for further details.

c. $V_{DD-A}$ = 2.8V, $V_{DD-D}$ = 1.3V, and $V_{DD-IO}$ = 1.8V for 15 fps in UXGA mode

d. $I_{DDS-SCCB}$ refers to SCCB-initiated Standby, while $I_{DDS-PWDN}$ refers to PWDN pad-initiated Standby

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**Table 7 AC Characteristics ($T_A$ = 25°C, $V_{DD-A}$ = 2.8V)**

| Symbol | Parameter | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| **ADC Parameters** | | | | | |
| B | Analog bandwidth | | 20 | | MHz |
| DLE | DC differential linearity error | | 0.5 | | LSB |
| ILE | DC integral linearity error | | 1 | | LSB |
| | Settling time for hardware reset | | | <1 | ms |
| | Settling time for software reset | | | <1 | ms |
| | Settling time for UXGA/SVGA mode change | | | <1 | ms |
| | Settling time for register setting | | | <300 | ms |

**Table 8 Timing Characteristics**

| Symbol | Parameter | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| **Oscillator and Clock Input** | | | | | |
| $f_{OSC}$ | Frequency (XVCLK) | 6 | 24 | | MHz |
| $t_r$, $t_f$ | Clock input rise/fall time | | | 5 | ns |
| | Clock input duty cycle | 45 | 50 | 55 | % |

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## Timing Specifications

**Figure 13 SCCB Interface Timing Diagram**

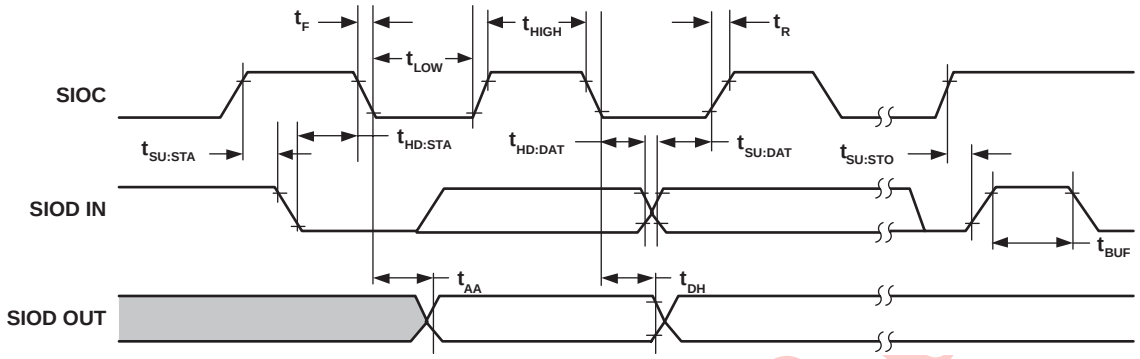



**Table 9 SCCB InterfaceTiming Specifications**

| Symbol | Parameter | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| $f_{SIO_C}$ | Clock Frequency | | | 400 | KHz |
| $t_{LOW}$ | Clock Low Period | 1.3 | | | µs |
| $t_{HIGH}$ | Clock High Period | 600 | | | ns |
| $t_{AA}$ | SIOC low to Data Out valid | 100 | | 900 | ns |
| $t_{BUF}$ | Bus free time before new START | 1.3 | | | µs |
| $t_{HD:STA}$ | START condition Hold time | 600 | | | ns |
| $t_{SU:STA}$ | START condition Setup time | 600 | | | ns |
| $t_{HD:DAT}$ | Data-in Hold time | 0 | | | µs |
| $t_{SU:DAT}$ | Data-in Setup time | 100 | | | ns |
| $t_{SU:STO}$ | STOP condition Setup time | 600 | | | ns |
| $t_R$, $t_F$ | SCCB Rise/Fall times | | | 300 | ns |
| $t_{DH}$ | Data-out Hold time | 50 | | | ns |

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**Figure 14 UXGA, SVGA, and CIF Line/Pixel Output Timing**

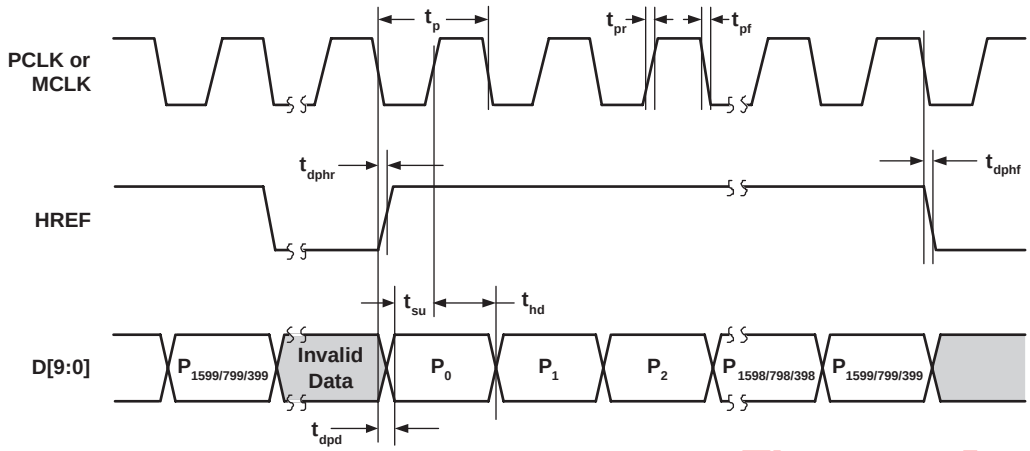



**Table 10 Pixel Timing Specifications**

| Symbol | Parameter | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| $t_p$ | PCLK period | | 27.78 | | ns |
| $t_{pr}$ | PCLK rising time | | 3.5 | | ns |
| $t_{pf}$ | PCLK falling time | | 2.2 | | ns |
| $t_{dphr}$ | PCLK negative edge to HREF rising edge | 0 | | 5 | ns |
| $t_{dphf}$ | PCLK negative edge to HREF negative edge | 0 | | 5 | ns |
| $t_{dpd}$ | PCLK negative edge to data output delay | 0 | | 5 | ns |
| $t_{su}$ | Data bus setup time | 15 | | | ns |
| $t_{hd}$ | Data bus hold time | 8 | | | ns |

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**Figure 15 UXGA Frame Timing**

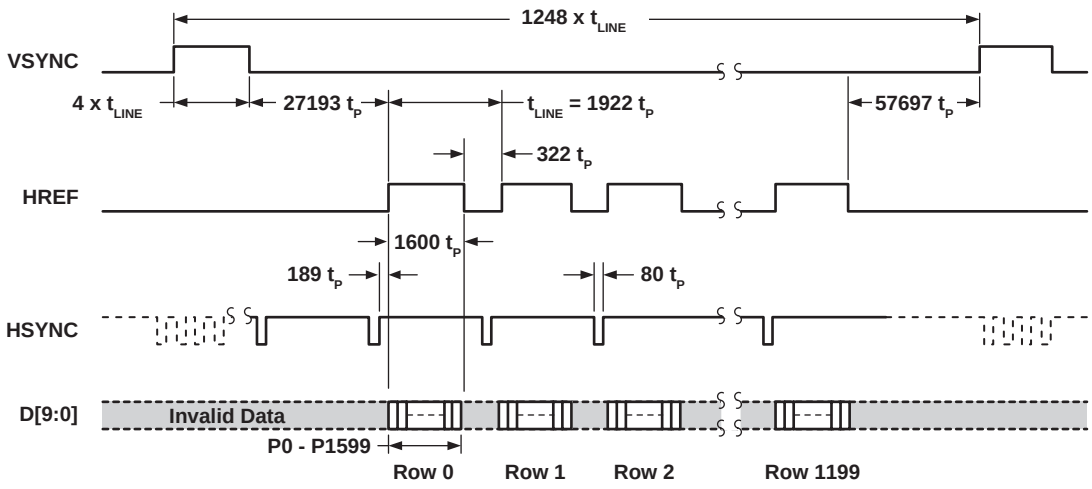



**Figure 16 SVGA Frame Timing**

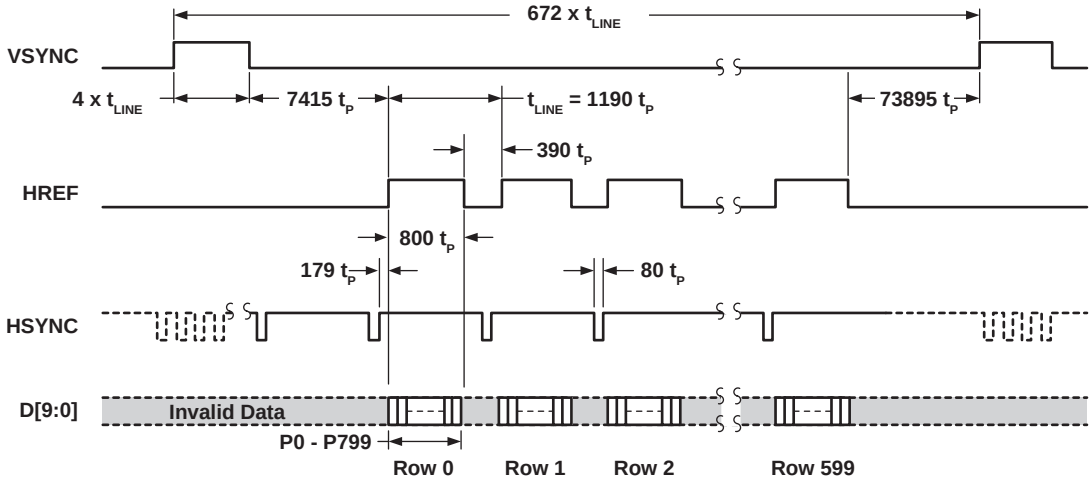



**Figure 17 CIF Mode Frame Timing**

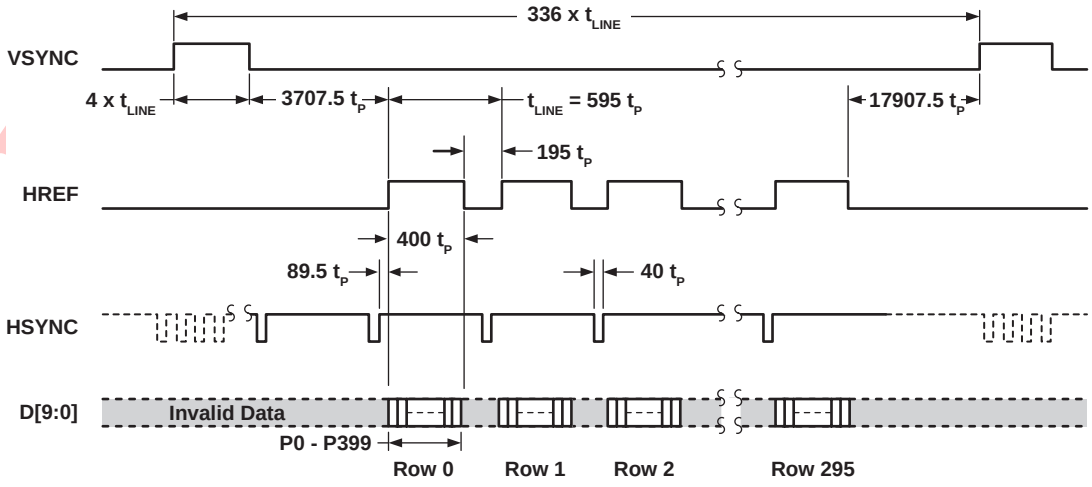


**Figure 18 Frame Exposure Mode Timing with EXPST_B Staying Low**

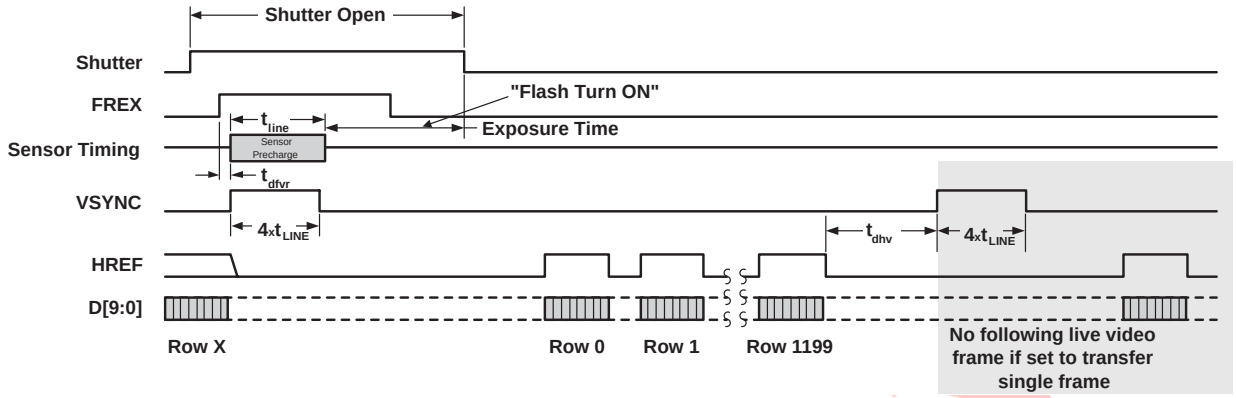



**Figure 19 Frame Exposure Mode Timing with EXPST_B Asserted**

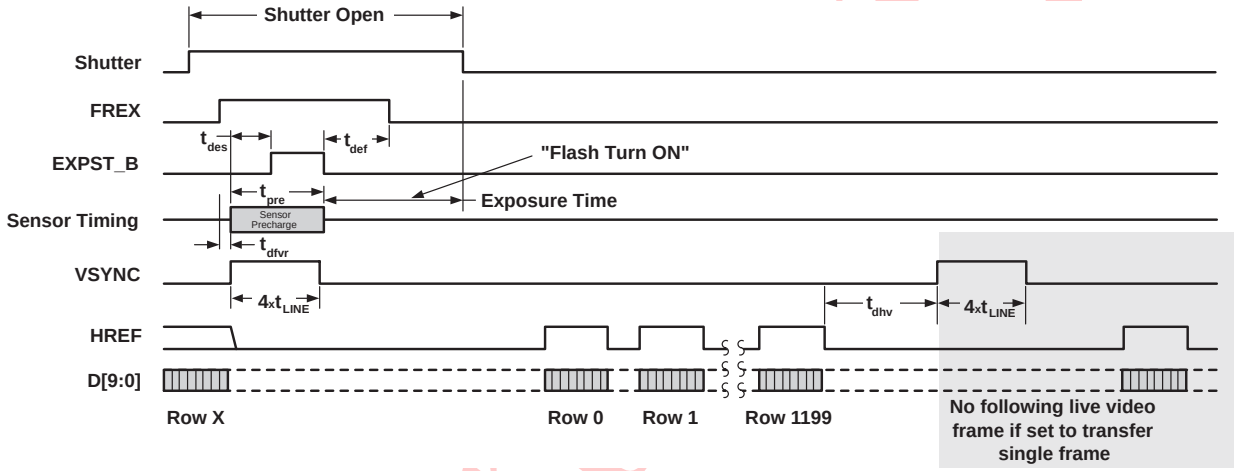


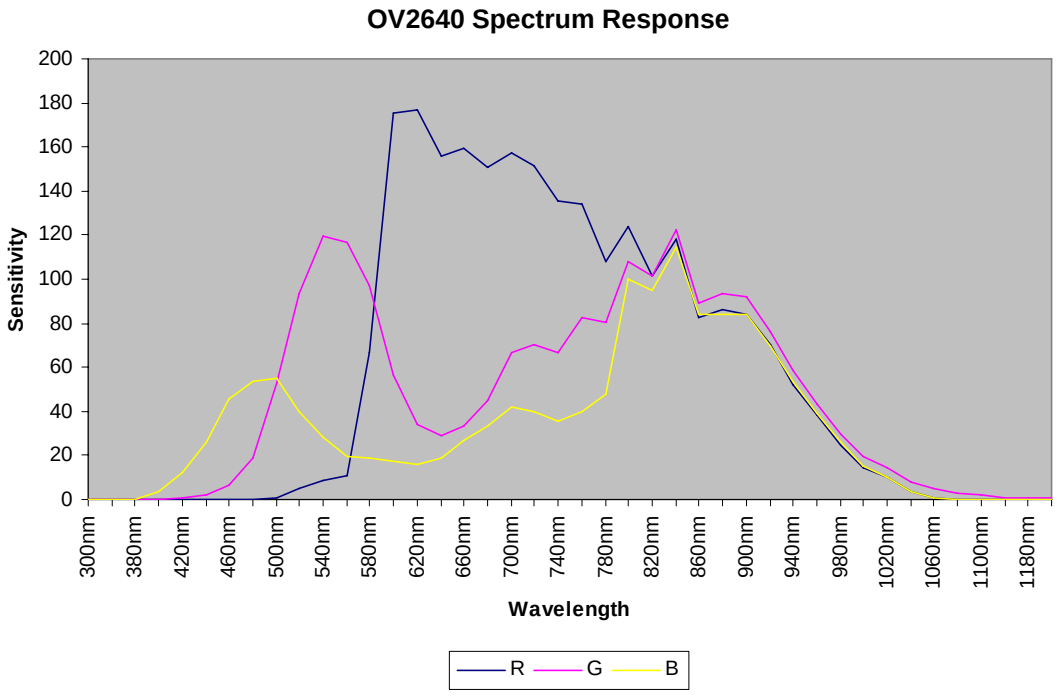

**Table 11 Frame Exposure Timing Specifications**

| Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|
| tline | | 1922 (UXGA) | | tp |
| tvs | | 4 | | tline |
| tdfvr | 8 | | 9 | tp |
| tdhv | | 38964 (UXGA) | | tp |
| tdhso | 0 | | | ns |
| tdef | 20 | | | tp |
| tdes | 8 | | 1900 (UXGA) | tp |

**NOTE** 1) FREX must stay high long enough to ensure the entire sensor has been reset.
2) Shutter must be closed no later then 3896 tp after VSYNC falling edge.

## OV2640 Light Response

**Figure 20 OV2640 Light Response**




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## Register Set

Table 12 and Table 13 provides a list and description of the Device Control registers contained in the OV2640. For all register Enable/Disable bits, ENABLE = 1 and DISABLE = 0. The device slave addresses are 60 for write and 61 for read.

There are two different sets of register banks. Register 0xFF controls which set is accessible. When register 0xFF=00, Table 12 is effective. When register 0xFF=01, Table 13 is effective.

**Table 12 Device Control Register List (when 0xFF = 00) (Sheet 1 of 4)**

| Address (Hex) | Register Name | Default (Hex) | R/W | Description |
|---|---|---|---|---|
| 00-04 | RSVD | XX | – | Reserved |
| 05 | R_BYPASS | 0x1 | RW | Bypass DSP<br>Bit[7:1]: Reserved<br>Bit[0]: Bypass DSP select<br>0: DSP<br>1: Bypass DSP, sensor out directly |
| 06-43 | RSVD | XX | – | Reserved |
| 44 | Qs | 0C | RW | Quantization Scale Factor |
| 45-4F | RSVD | XX | – | Reserved |
| 50 | CTRLI[7:0] | 00 | RW | Bit[7]: LP_DP<br>Bit[6]: Round<br>Bit[5:3]: V_DIVIDER<br>Bit[2:0]: H_DIVIDER |
| 51 | HSIZE[7:0] | 40 | RW | H_SIZE[7:0] (real/4) |
| 52 | VSIZE[7:0] | F0 | RW | V_SIZE[7:0] (real/4) |
| 53 | XOFFL[7:0] | 00 | RW | OFFSET_X[7:0] |
| 54 | YOFFL[7:0] | 00 | RW | OFFSET_Y[7:0] |
| 55 | VHYX[7:0] | 08 | RW | Bit[7]: V_SIZE[8]<br>Bit[6:4]: OFFSET_Y[10:8]<br>Bit[3]: H_SIZE[8]<br>Bit[2:0]: OFFSET_X[10:8] |
| 56 | DPRP[7:0] | 00 | RW | Bit[7:4]: DP_SELY<br>Bit[3:0]: DP_SELX |
| 57 | TEST[3:0] | 00 | RW | Bit[7]: H_SIZE[9]<br>Bit[6:0]: Reserved |
| 5A | ZMOW[7:0] | 58 | RW | OUTW[7:0] (real/4) |
| 5B | ZMOH[7:0] | 48 | RW | OUTH[7:0] (real/4) |
| 5C | ZMHH[1:0] | 00 | RW | Bit[7:4]: ZMSPD (zoom speed)<br>Bit[2]: OUTH[8]<br>Bit[1:0]: OUTW[9:8] |
| 5D-7B | RSVD | XX | – | Reserved |
| 7C | BPADDR[3:0] | 00 | RW | SDE Indirect Register Access: Address |

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**Table 12 Device Control Register List (when 0xFF = 00) (Sheet 2 of 4)**

| Address (Hex) | Register Name | Default (Hex) | R/W | Description |
|---|---|---|---|---|
| 7D | BPDATA[7:0] | 00 | RW | SDE Indirect Register Access: Data |
| 7E-85 | RSVD | XX | – | Reserved |
| 86 | CTRL2 | 0D | RW | Module Enable<br>Bit[7:6]: Reserved<br>Bit[5]: DCW<br>Bit[4]: SDE<br>Bit[3]: UV_ADJ<br>Bit[2]: UV_AVG<br>Bit[1]: Reserved<br>Bit[0]: CMX |
| 87 | CTRL3 | 50 | RW | Module Enable<br>Bit[7]: BPC<br>Bit[6]: WPC<br>Bit[5:0]: Reserved |
| 88-8B | RSVD | XX | – | Reserved |
| 8C | SIZEL[5:0] | 00 | RW | {HSIZE[11], HSIZE[2:0], VSIZE[2:0]} |
| 8D-BF | RSVD | XX | – | Reserved |
| C0 | HSIZE8[7:0] | 80 | RW | Image Horizontal Size HSIZE[10:3] |
| C1 | VSIZE8[7:0] | 60 | RW | Image Vertical Size VSIZE[10:3] |
| C2 | CTRL0 | 0C | RW | Module Enable<br>Bit[7]: AEC_EN<br>Bit[6]: AEC_SEL<br>Bit[5]: STAT_SEL<br>Bit[4]: VFIRST<br>Bit[3]: YUV422<br>Bit[2]: YUV_EN<br>Bit[1]: RGB_EN<br>Bit[0]: RAW_EN |
| C3 | CTRL1 | FF | RW | Module Enable<br>Bit[7]: CIP<br>Bit[6]: DMY<br>Bit[5]: RAW_GMA<br>Bit[4]: DG<br>Bit[3]: AWB<br>Bit[2]: AWB_GAIN<br>Bit[1]: LENC<br>Bit[0]: PRE |
| C4-D2 | RSVD | XX | – | Reserved |

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**Table 12 Device Control Register List (when 0xFF = 00) (Sheet 3 of 4)**

| Address (Hex) | Register Name | Default (Hex) | R/W | Description |
|---|---|---|---|---|
| D3 | R_DVP_SP | 82 | RW | Bit[7]: Auto mode<br>Bit[6:0]: DVP output speed control<br>DVP PCLK = sysclk (48)/[6:0] (YUV0);<br>= sysclk (48)/(2*[6:0]) (RAW) |
| D4-D9 | RSVD | XX | – | Reserved |
| DA | IMAGE_MODE | 00 | | Image Output Format Select<br>Bit[7]: Reserved<br>Bit[6]: Y8 enable for DVP<br>Bit[5]: Reserved<br>Bit[4]: JPEG output enable<br>0: Non-compressed<br>1: JPEG output<br>Bit[3:2]: DVP output format<br>00: YUV422<br>01: RAW10 (DVP)<br>10: RGB565<br>11: Reserved<br>Bit[1]: HREF timing select in DVP JPEG output mode<br>0: HREF is same as sensor<br>1: HREF = VSYNC<br>Bit[0]: Byte swap enable for DVP<br>0: High byte first YUYV (C2[4]=0) YVYU (C2[4] = 1)<br>1: Low byte first UYVY (C2[4] =0) VYUY (C2[4] =1) |
| DB-DF | RSVD | XX | – | Reserved |
| E0 | RESET | 04 | RW | Reset<br>Bit[7]: Reserved<br>Bit[6]: Microcontroller<br>Bit[5]: SCCB<br>Bit[4]: JPEG<br>Bit[3]: Reserved<br>Bit[2]: DVP<br>Bit[1]: IPU<br>Bit[0]: CIF |
| E1-EF | RSVD | XX | – | Reserved |
| F0 | MS_SP | 04 | RW | SCCB Master Speed |
| F1-F6 | RSVD | XX | – | Reserved |
| F7 | SS_ID | | RW | SCCB Slave ID |

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**Table 12 Device Control Register List (when 0xFF = 00) (Sheet 4 of 4)**

| Address (Hex) | Register Name | Default (Hex) | R/W | Description |
|---|---|---|---|---|
| F8 | SS_CTRL | 01 | RW | SCCB Slave Control<br>Bit[7:6]: Reserved<br>Bit[5]: Address auto-increase enable<br>Bit[4]: Reserved<br>Bit[3]: SCCB enable<br>Bit[2]: Delay SCCB master clock<br>Bit[1]: Enable SCCB master access<br>Bit[0]: Enable sensor pass through access |
| F9 | MC_BIST | | RW | Bit[7]: Microcontroller Reset<br>Bit[6]: Boot ROM select<br>Bit[5]: R/W 1 error for 12K-byte memory<br>Bit[4]: R/W 0 error for 12K-byte memory<br>Bit[3]: R/W 1 error for 512-byte memory<br>Bit[2]: R/W 0 error for 512-byte memory<br>Bit[1]: BIST busy bit for read; One-shot reset of microcontroller for write<br>Bit[0]: Launch BIST |
| FA | MC_AL | | RW | Program Memory Pointer Address Low Byte |
| FB | MC_AH | | RW | Program Memory Pointer Address High Byte |
| FC | MC_D | 80 | RW | Program Memory Pointer Access Address<br>Boundary of register address to separate DSP and sensor register |
| FD | P_CMD | 00 | RW | SCCB Protocol Command Register |
| FE | P_STATUS | 00 | RW | SCCB Protocol Status Register |
| FF | RA_DLMT | 7F | RW | Register Bank Select<br>Bit[7:1]: Reserved<br>Bit[0]: Register bank select<br>0: DSP address<br>1: Sensor address |
| **NOTE:** All other registers are factory-reserved. Please contact OmniVision Technologies for reference register settings. | | | | |

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**Table 13 Device Control Register List (when 0xFF = 01) (Sheet 1 of 7)**

| Address (Hex) | Register Name | Default (Hex) | R/W | Description |
|---|---|---|---|---|
| 00 | GAIN | 00 | RW | AGC Gain Control LSBs<br>Bit[7:0]: Gain setting<br>• Range: 1x to 32x<br>Gain = (Bit[7]+1) x (Bit[6]+1) x (Bit[5]+1) x (Bit[4]+1) x (1+Bit[3:0]/16)<br>***Note:*** *Set COM8[2] = 0 to disable AGC.* |
| 01-02 | RSVD | XX | – | Reserved |
| 03 | COM1 | 0F (UXGA)<br>0A (SVGA),<br>06 (CIF) | RW | Common Control 1<br>Bit[7:6]: Dummy frame control<br>00: Reserved<br>01: Allow 1 dummy frame<br>10: Allow 3 dummy frames<br>11: Allow 7 dummy frames<br>Bit[5:4]: Reserved<br>Bit[3:2]: Vertical window end line control 2 LSBs (8 MSBs in VEND[7:0] (0x1A))<br>Bit[1:0]: Vertical window start line control 2 LSBs (8 MSBs in VSTRT[7:0] (0x19)) |
| 04 | REG04 | 20 | RW | Register 04<br>Bit[7]: Horizontal mirror<br>Bit[6]: Vertical flip<br>Bit[4]: VREF bit[0]<br>Bit[3]: HREF bit[0]<br>Bit[2]: Reserved<br>Bit[1:0]: AEC[1:0] (AEC[15:10] is in register REG45[5:0] (0x45), AEC[9:2] is in register AEC[7:0] (0x10)) |
| 05-07 | RSVD | XX | – | Reserved |
| 08 | REG08 | 40 | RW | Frame Exposure One-pin Control Pre-charge Row Number |
| 09 | COM2 | 00 | RW | Common Control 2<br>Bit[7:5]: Reserved<br>Bit[4]: Standby mode enable<br>0: Normal mode<br>1: Standby mode<br>Bit[3]: Reserved<br>Bit[2]: Pin PWDN/RESETB used as SLVS/SLHS<br>Bit[1:0]: Output drive select<br>00: 1x capability<br>01: 3x capability<br>10: 2x capability<br>11: 4x capability |
| 0A | PIDH | 26 | R | Product ID Number MSB (Read only) |
| 0B | PIDL | 41 | R | Product ID Number LSB (Read only) |

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**Table 13 Device Control Register List (when 0xFF = 01) (Sheet 2 of 7)**

| Address (Hex) | Register Name | Default (Hex) | R/W | Description |
|---|---|---|---|---|
| 0C | COM3 | 38 | RW | Common Control 3<br>Bit[7:3]: Reserved<br>Bit[2]: Set banding manually<br>0: 60 Hz<br>1: 50 Hz<br>Bit[1]: Auto set banding<br>Bit[0]: Snapshot option<br>0: Enable live video output after snapshot sequence<br>1: Output single frame only |
| 0D | COM4 | 07 | RW | Common Control 4<br>Bit[7:3]: Reserved<br>Bit[2]: Clock output power-down pin status<br>0: Tri-state data output pin upon power-down<br>1: Data output pin hold at last state before power-down<br>Bit[1:0]: Reserved |
| 0E-0F | RSVD | XX | – | Reserved |
| 10 | AEC | 33 | RW | Automatic Exposure Control 8 bits for AEC[9:2] (AEC[15:10] is in register REG45[5:0] (0x45), AEC[1:0] is in register REG04[1:0] (0x04))<br>AEC[15:0]: Exposure time<br><br>$T_{EX} = t_{LINE}$ x AEC[15:0]<br><br>***Note:*** *The maximum exposure time is 1 frame period even if TEX is longer than 1 frame period.* |
| 11 | CLKRC | 00 | RW | Clock Rate Control<br>Bit[7]: Internal frequency doublers ON/OFF selection<br>0: OFF<br>1: ON<br>Bit[6]: Reserved<br>Bit[5:0]: Clock divider<br><br>CLK = XVCLK/(decimal value of CLKRC[5:0] + 1) |

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**Table 13 Device Control Register List (when 0xFF = 01) (Sheet 3 of 7)**

| Address (Hex) | Register Name | Default (Hex) | R/W | Description |
|---|---|---|---|---|
| 12 | COM7 | 00 | RW | Common Control 7<br>Bit[7]: SRST<br>1: Initiates system reset. All registers are set to factory default values after which the chip resumes normal operation<br>Bit[6:4]: Resolution selection<br>000: UXGA (full size) mode<br>001: CIF mode<br>100: SVGA mode<br>Bit[3]: Reserved<br>Bit[2]: Zoom mode<br>Bit[1]: Color bar test pattern<br>0: OFF<br>1: ON<br>Bit[0]: Reserved |
| 13 | COM8 | C7 | RW | Common Control 8<br>Bit[7:6]: Reserved<br>Bit[5]: Banding filter selection<br>0: OFF<br>1: ON, set minimum exposure time to 1/120s<br>Bit[4:3]: Reserved<br>Bit[2]: AGC auto/manual control selection<br>0: Manual<br>1: Auto<br>Bit[1]: Reserved<br>Bit[0]: Exposure control<br>0: Manual<br>1: Auto |
| 14 | COM9 | 50 | RW | Common Control 9<br>Bit[7:5]: AGC gain ceiling, GH[2:0]<br>000: 2x<br>001: 4x<br>010: 8x<br>011: 16x<br>100: 32x<br>101: 64x<br>11x: 128x<br>Bit[4:0]: Reserved |

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**Table 13 Device Control Register List (when 0xFF = 01) (Sheet 4 of 7)**

| Address (Hex) | Register Name | Default (Hex) | R/W | Description |
|---|---|---|---|---|
| 15 | COM10 | 00 | RW | Common Control 10 (if Bypass DSP is selected)<br>Bit[7]: CHSYNC pin output swap<br>0: CHSYNC<br>1: HREF<br>Bit[6]: HREF pin output swap<br>0: HREF<br>1: CHSYNC<br>Bit[5]: PCLK output selection<br>0: PCLK always output<br>1: PCLK output qualified by HREF<br>Bit[4]: PCLK edge selection<br>0: Data is updated at the falling edge of PCLK (user can latch data at the next rising edge of PCLK)<br>1: Data is updated at the rising edge of PCLK (user can latch data at the next falling edge of PCLK)<br>Bit[3]: HREF output polarity<br>0: Output positive HREF<br>1: Output negative HREF, HREF negative for data valid<br>Bit[2]: Reserved<br>Bit[1]: VSYNC polarity<br>0: Positive<br>1: Negative<br>Bit[0]: HSYNC polarity<br>0: Positive<br>1: Negative |
| 16 | RSVD | XX | – | Reserved |
| 17 | HREFST | 11 | RW | Horizontal Window Start MSB 8 bits (3 LSBs in REG32[2:0] (0x32))<br>Bit[10:0]: Selects the start of the horizontal window, each LSB represents two pixels |
| 18 | HREFEND | 75 (UXGA), 43 (SVGA, CIF) | RW | Horizontal Window End MSB 8 bits (3 LSBs in REG32[5:3] (0x32))<br>Bit[10:0]: Selects the end of the horizontal window, each LSB represents two pixels |
| 19 | VSTRT | 01 (UXGA), 00 (SVGA, CIF) | RW | Vertical Window Line Start MSB 8 bits (2 LSBs in COM1[1:0] (0x03))<br>Bit[9:0]: Selects the start of the vertical window, each LSB represents two scan lines. |
| 1A | VEND | 97 | RW | Vertical Window Line End MSB 8 bits (2 LSBs in COM1[3:2] (0x03))<br>Bit[9:0]: Selects the end of the vertical window, each LSB represents two scan lines. |
| 1B | RSVD | XX | – | Reserved |
| 1C | MIDH | 7F | R | Manufacturer ID Byte – High (Read only = 0x7F) |
| 1D | MIDL | A2 | R | Manufacturer ID Byte – Low (Read only = 0xA2) |
| 1E-23 | RSVD | XX | – | Reserved |

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**Table 13 Device Control Register List (when 0xFF = 01) (Sheet 5 of 7)**

| Address (Hex) | Register Name | Default (Hex) | R/W | Description |
|---|---|---|---|---|
| 24 | AEW | 78 | RW | Luminance Signal High Range for AEC/AGC Operation<br>AEC/AGC values will decrease in auto mode when average luminance is greater than AEW[7:0] |
| 25 | AEB | 68 | RW | Luminance Signal Low Range for AEC/AGC Operation<br>AEC/AGC values will increase in auto mode when average luminance is less than AEB[7:0] |
| 26 | VV | D4 | RW | Fast Mode Large Step Range Threshold - effective only in AEC/AGC fast mode (COM8[7] = 1)<br>Bit[7:4]: High threshold<br>Bit[3:0]:Low threshold<br>***Note:*** *AEC/AGC may change in larger steps when luminance average is greater than VV[7:4] or less than VV[3:0].* |
| 27-29 | RSVD | XX | – | Reserved |
| 2A | REG2A | 00 | RW | Register 2A<br>Bit[7:4]: Line interval adjust value 4 MSBs (LSBs in FRARL[7:0] (0x2B))<br>Bit[3:2]: HSYNC timing end point adjustment MSB 2 bits (LSBs in register HEDY[7:0] (0x31))<br>Bit[1:0]: HSYNC timing start point adjustment MSB 2 bits (LSBs in register HSDY[7:0] (0x30)) |
| 2B | FRARL | 00 | RW | Line Interval Adjustment Value LSB 8 bits (MSBs in REG2A[7:4] (0x2A))<br>The frame rate will be adjusted by changing the line interval. Each LSB will add 1/1922 $T_{frame}$ in UXGA and 1/1190 $T_{frame}$ in SVGA mode to te frame period. |
| 2C | RSVD | XX | – | Reserved |
| 2D | ADDVSL | 00 | RW | VSYNC Pulse Width LSB 8 bits<br>Bit[7:0]: Line periods added to VSYNC width. Default VSYNC output width is 4 x $t_{line}$. Each LSB count will add 1 x $t_{line}$ to the VSYNC active period. |
| 2E | ADDVSH | 00 | RW | VSYNC Pulse Width MSB 8 bits<br>Bit[7:0]: Line periods added to VSYNC width. Default VSYNC output width is 4 x $t_{line}$. Each MSB count will add 256 x $t_{line}$ to the VSYNC active period. |
| 2F | YAVG | 00 | RW | Luminance Average (this register will auto update)<br>Average Luminance is calculated from the B/Gb/Gr/R channel average as follows:<br>B/Gb/Gr/R channel average =<br>(BAVG[7:0] + (2 x GbAVG[7:0]) + RAVG[7:0]) x 0.25 |
| 30 | HSDY | 08 | RW | HSYNC Position and Width, Start Point LSB 8 bits<br>This register and REG2A[1:0] (0x2A) define HSYNC start position, each LSB will shift HSYNC start by 2 pixel period |

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**Table 13 Device Control Register List (when 0xFF = 01) (Sheet 6 of 7)**

| Address (Hex) | Register Name | Default (Hex) | R/W | Description |
|---|---|---|---|---|
| 31 | HEDY | 30 | RW | HSYNC Position and Width, End Point LSB 8 bits<br>This register and REG2A[3:2] (0x2A) define HSYNC end position, each LSB will shift HSYNC end by 2 pixel period |
| 32 | REG32 | 36 (UXGA), 09 (SVGA, CIF) | RW | Common Control 32<br>Bit[7:6]: Pixel clock divide option<br>00: No effect on PCLK<br>01: No effect on PCLK<br>10: PCLK frequency divide by 2<br>11: PCLK frequency divide by 4<br>Bit[5:3]: Horizontal window end position 3 LSBs (8 MSBs in register HREFEND[7:0] (0x18))<br>Bit[2:0]: Horizontal window start position 3 LSBs (8 MSBs in register HREFST[7:0] (0x17)) |
| 33 | RSVD | XX | – | Reserved |
| 34 | ARCOM2 | 20 | RW | Bit[7:3]: Reserved<br>Bit[2]: Zoom window horizontal start point<br>Bit[1:0]: Reserved |
| 35-44 | RSVD | XX | – | Reserved |
| 45 | REG45 | 00 | RW | Register 45<br>Bit[7:6]: AGC[9:8], AGC highest gain control<br>Bit[5:0]: AEC[15:10], AEC MSBs |
| 46 | FLL | 00 | RW | Frame Length Adjustment LSBs<br>Each bit will add 1 horizontal line timing in frame |
| 47 | FLH | 00 | RW | Frame Length Adjustment MSBs<br>Each bit will add 256 horizontal lines timing in frame |
| 48 | COM19 | 00 | RW | Common Control 19<br>Bit[7:2]: Reserved<br>Bit[1:0]: Zoom mode vertical window start point 2 LSBs |
| 49 | ZOOMS | 00 | RW | Zoom Mode Vertical Window Start Point 8 MSBs |
| 4A | RSVD | XX | – | Reserved |
| 4B | COM22 | 20 | RW | Common Control 22<br>Bit[7:0]: Flash light control |
| 4C-4D | RSVD | XX | – | Reserved |
| 4E | COM25 | 00 | RW | Common Control 25 - reserved for banding<br>Bit[7:6]: 50Hz Banding AEC 2 MSBs<br>Bit[5:4]: 60HZ Banding AEC 2 MSBs<br>Bit[3:0]: Reserved |
| 4F | BD50 | CA | RW | 50Hz Banding AEC 8 LSBs |
| 50 | BD60 | A8 | RW | 60Hz Banding AEC 8 LSBs |
| 51-5C | RSVD | XX | – | Reserved |

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**Table 13 Device Control Register List (when 0xFF = 01) (Sheet 7 of 7)**

| Address (Hex) | Register Name | Default (Hex) | R/W | Description |
|---|---|---|---|---|
| 5D | REG5D | 00 | RW | Register 5D<br>Bit[7:0]: AVGsel[7:0], 16-zone average weight option |
| 5E | REG5E | 00 | RW | Register 5E<br>Bit[7:0]: AVGsel[15:8], 16-zone average weight option |
| 5F | REG5F | 00 | RW | Register 5F<br>Bit[7:0]: AVGsel[23:16], 16-zone average weight option |
| 60 | REG60 | 00 | RW | Register 60<br>Bit[7:0]: AVGsel[31:24], 16-zone average weight option |
| 61 | HISTO_LOW | 80 | RW | Histogram Algorithm Low Level |
| 62 | HISTO_HIGH | 90 | RW | Histogram Algorithm High Level |
| 63-7E | RSVD | XX | – | Reserved |
| **NOTE:** All other registers are factory-reserved. Please contact OmniVision Technologies for reference register settings. | | | | |

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## Package Specifications

The OV2640 uses a 38-ball Chip Scale Package 2 (CSP2). Refer to Figure 11 for package information, Table 9 for package dimensions and Figure 12 for the array center on the chip.

**Note:** For OVT devices that are lead-free, all part marking letters are lower case. Underlining the last digit of the lot number indicates CSP2 is used.

**Figure 21 OV2640 Package Specifications**

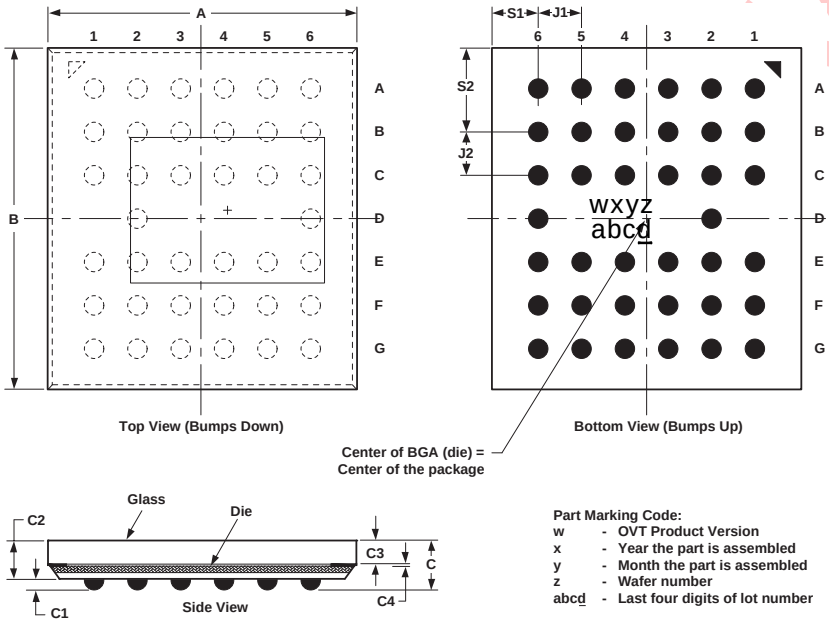


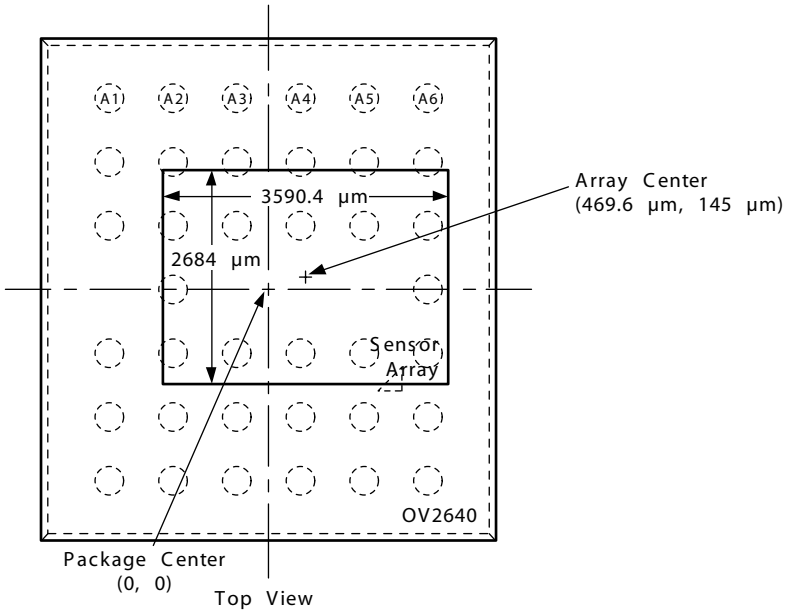

**Table 14 OV2640 Package Dimensions**

| Parameter | Symbol | Minimum | Nominal | Maximum | Unit |
|---|---|---|---|---|---|
| Package Body Dimension X | A | 5700 | 5725 | 5750 | µm |
| Package Body Dimension Y | B | 6260 | 6285 | 6310 | µm |
| Package Height | C | 845 | 905 | 965 | µm |
| Ball Height | C1 | 150 | 180 | 210 | µm |
| Package Body Thickness | C2 | 680 | 725 | 770 | µm |
| Cover Glass Thickness | C3 | 375 | 400 | 425 | µm |
| Airgap Between Cover Glass and Sensor | C4 | 30 | 45 | 60 | µm |
| Ball Diameter | D | 320 | 350 | 380 | µm |
| Total Pin Count | N | | 38 (1 NC) | | |
| Pin Count X-axis | N1 | | 6 | | |
| Pin Count Y-axis | N2 | | 7 | | |
| Pins Pitch X-axis | J1 | | 800 | | µm |
| Pins Pitch Y-axis | J2 | | 800 | | µm |
| Edge-to-Pin Center Distance Analog X | S1 | 833 | 863 | 893 | µm |
| Edge-to-Pin Center Distance Analog Y | S2 | 713 | 743 | 773 | µm |

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## Sensor Array Center

**Figure 22 OV2640 Sensor Array Center**




NOTES: 1. This drawing is not to scale and is for reference only.

2. As most optical assemblies invert and mirror the image, the chip is typically mounted with pins A1 to A6 oriented down on the PCB.

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## IR Reflow Ramp Rate Requirements

### OV2640 Lead-Free Packaged Devices

**Note:** For OVT devices that are lead-free, all part marking letters are lower case

**Figure 23 IR Reflow Ramp Rate Requirements**

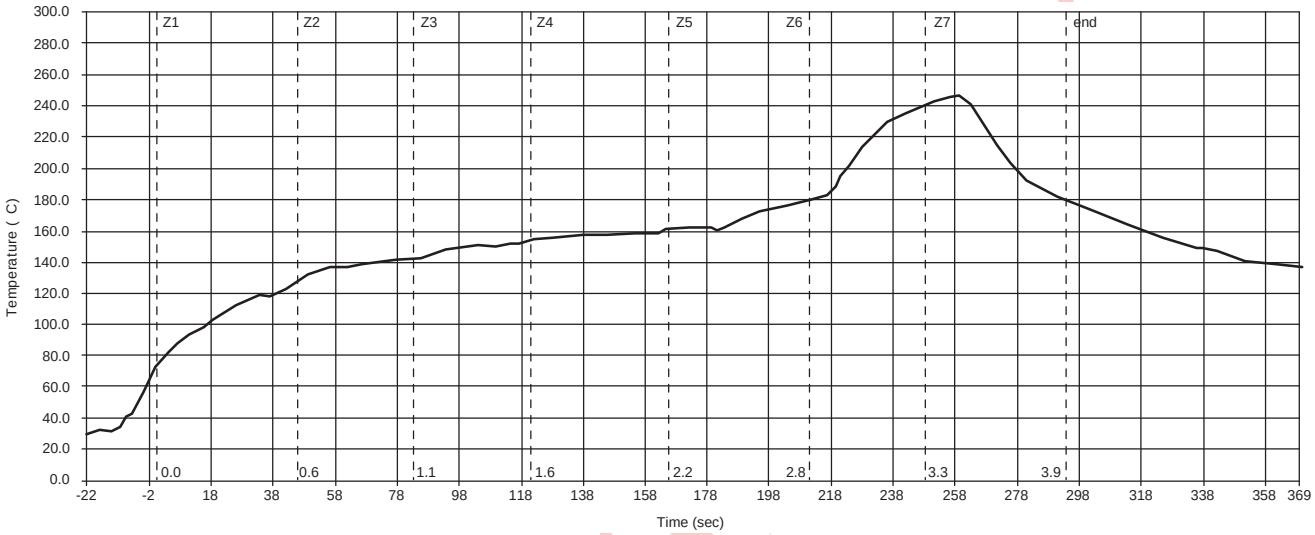



**Table 15 Reflow Conditions**

| Condition | Exposure |
|---|---|
| Average Ramp-up Rate (30°C to 217°C) | Less than 3°C per second |
| > 100°C | Between 330 - 600 seconds |
| > 150°C | At least 210 seconds |
| > 217°C | At least 30 seconds (30 ~ 120 seconds) |
| Peak Temperature | 245°C |
| Cool-down Rate (Peak to 50°C) | Less than 6°C per second |
| Time from 30°C to 245°C | No greater than 390 seconds |

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## Note:

- *All information shown herein is current as of the revision and publication date. Please refer to the OmniVision web site (http://www.ovt.com) to obtain the current versions of all documentation.*

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# REVISION CHANGE LIST

**Document Title:** OV2640 Datasheet **Version:** 1.0

# DESCRIPTION OF CHANGES

Initial Release

# REVISION CHANGE LIST

**Document Title:** OV2640 Datasheet **Version:** 1.01

# DESCRIPTION OF CHANGES

The following changes were made to version 1.0:

- Under Key Specifications on page 1, changed specification for Core Power Supply from "1.2VDC ± 10%" to "1.2VDC ± 5%"
- Under Key Specifications on page 1, changed specification for Analog Power Supply from "2.8VDC ± 10%" to "2.5 ~ 3.0VDC"
- Under Key Specifications on page 1, changed specification for I/O Power Supply from "1.8V to 3.3V" to "1.7V to 3.3V"
- On pages 17 to 20, changed title of Table 12 from "Device Control Register (for 0x00 ~ 0xFF at 0xF8 = 00 and 0xFF = 00)" to "Device Control Register (when 0xFF = 00)"
- On pages 21 to 27, changed title of Table 13 from "Device Control Register (for 0x00 ~ 0x7E at 0xF8 = 01 and 0xFF = 7F)" to "Device Control Register (when 0xFF = 01)"
- In Table 12 on pages 18, changed description of register CTRL3 (0x87) from:

  Module Enable

  Bit[7:6]: Reserved
  Bit[5]: DCW
  Bit[4]: SDE
  Bit[3]: UV_ADJ
  Bit[2]: UV_AVG
  Bit[1]: Reserved
  Bit[0]: CMX

  to

  Module Enable

  Bit[7]: BPC
  Bit[6]: WPC
  Bit[5:0]: Reserved

- In Table 15 on page 30, changed specification for Peak Temperature from "Greater than 245°C" to "245°C"

# REVISION CHANGE LIST

**Document Title:** OV2640 Datasheet **Version:** 1.1

## DESCRIPTION OF CHANGES

The following changes were made to version 1.01:

- Under Features on page 1, changed bulleted item from “Supports image sizes: UXGA, SVGA, and any size scaling down from SVGA to 40x30” to “Supports image sizes: UXGA, SXGA, SVGA, and any size scaling down from SXGA to 40x30”
- Under Key Specifications on page 1, deleted specifications for SVGA and CIF Array Size
- Under Key Specifications on page 1, changed Standby Power Requirements specification to “TBD”
- Under Key Specifications on page 1, changed specification for Chief Ray Angle from “TBD” to “25° non-linear”
- Under Key Specifications on page 1, changed specification for Well Capacity from “TBD” to “12 Ke”
- Under Electrical Characteristics on page 10, changed title of Table 6 from “DC Characteristics (-20°C < $T_A$ < 70°C)” to “DC Characteristics (-30°C < $T_A$ < 70°C)”
- In Table 6 on page 10, changed specification for Typ Standby Current from “10” to “TBD”
- In Table 6 on page 10, changed specification for Max Input voltage LOW ($V_{IL}$) from “0.8” to “0.54”
- In Table 6 on page 10, changed specification for Min Input voltage HIGH ($V_{IH}$) from “2” to “1.26”
- In Table 6 on page 10, changed subtitle “Digital Outputs (standard loading 25 pF, 1.2 KΩ to 2.8V)” to “Digital Outputs (standard loading 25 pF)”
- In Table 6 on page 10, changed specification for Min Output voltage HIGH ($V_{OH}$) from “2.2” to “1.62”
- In Table 6 on page 10, changed specification for Max Output voltage LOW ($V_{OL}$) from “0.6” to “0.18”
- In Table 6 on page 10, changed specification for Serial Interface Inputs Max SIO_C and SIO_D ($V_{IL}$) from “1” to “0.54”
- In Table 6 on page 10, changed specification for Serial Interface Inputs Min, Typ, and Max SIO_C and SIO_D ($V_{IH}$) from “2.5, 2.8, and VDD-IO + 0.5” to “1.26, 1.8, and 2.3”, respectively
- In Table 6 on page 10, changed table footnote b from “...$V_{DD-IO}$ = 2.8V” to “...$V_{DD-IO}$ = 1.8V”

# DESCRIPTION OF CHANGES (CONTINUED)

- In Figure 21 on page 28, changed callout C3 to measure from thickness of glass and added callout C4 to measure airgap from glass to die.
- In Table 14 on page 28, changed C3 parameter name from “Thickness of Glass Surface to Wafer” to “Cover Glass Thickness”
- In Table 14 on page 28, changed C3 Minimum, Nominal, and Maximum specifications from “425, 445, and 465” to “375, 400, and 425”
- In Table 14 on page 28, added C4 parameter, Airgap Between Cover Glass and Sensor, and Minimum, Nominal, and Maximum specifications “30, 45, and 60”, respectively

# REVISION CHANGE LIST

**Document Title:** OV2640 Datasheet **Version:** 1.2

# DESCRIPTION OF CHANGES

The following changes were made to version 1.1:

- Under Key Specifications on page 1, changed Active Power Requirements specification to "TBD" to "125 mW (for 15 fps, UXGA YUV mode)" and "140 mW (for 15 fps, UXGA compressed mode)"
- Under Key Specifications on page 1, changed Standby Power Requirements specification to "TBD" to "600 µA"
- Under Key Specifications on page 1, deleted Preview (CIF) Power Requirements specification
- In Table 6 on page 10, changed specification for Typ Active (Operating) Current ($I_{DDA-A}$) from "TBD" to "30"
- In Table 6 on page 10, changed specification for Typ Active (Operating) Current ($I_{DDA-D}$) from "TBD" to "25 (YUV)" and "35 (Compressed)"
- In Table 6 on page 10, changed specification for Typ Active (Operating) Current ($I_{DDA-IO}$) from "TBD" to "6"
- IIn Table 6 on page 10, changed specification for Typ Standby Current from "10" to "600"
- In Table 6 on page 10, changed table footnote b from "...$V_{DD-IO}$ = 1.8V" to "...$V_{DD-IO}$ = 1.8V for 15 fps in UXGA mode"

# REVISION CHANGE LIST

**Document Title:** OV2640 Datasheet **Version:** 1.21

# DESCRIPTION OF CHANGES

The following changes were made to version 1.2:

- In Figure 1 on page 21, corrected the bottom view of the package by correcting the column numbers corresponding to the ball locations from (left to right) “1”, “2”, “3”, “4”, “5”, and “6” to (left to right) “6”, “5”, “4”, “3”, “2”, and “1”, respectively

# REVISION CHANGE LIST

**Document Title:** OV2640 Datasheet **Version:** 1.3

# DESCRIPTION OF CHANGES

The following changes were made to version 1.21:

- In Table 1 on page 8, made the following changes/corrections:
  - Corrected pin type of pin A1 from Power to Ground
  - Corrected pin type of pin A2 from I/O to Input and added “Note: There is no internal pull-up/pull-down resistor”
  - Corrected pin type of pin A3 from Power to Ground
  - Corrected pin type of pin A4 from Power to Ground
  - Corrected pin type of pin A5 from I/O to Reference
  - Added “Default: Input” and “Note: There is no internal pull-up/pull-down resistor” to description of pin A6
  - Corrected pin type of pin B2 from Power to Input and added “Note: There is no internal pull-up/pull-down resistor”
  - Corrected pin type of pin B3 from Input to Power
  - Corrected pin type of pin B4 from I/O to Power
  - Corrected pin type of pin B5 from Input to Power
  - Corrected pin type of pin B6 from I/O to Input and “Note: There is an internal pull-down resistor”
  - Added “Default: Input” and “Note: There is no internal pull-up/pull-down resistor” to description of pin C3
  - Added “Note: There is no internal pull-up/pull-down resistor” to description of pin C4
  - Added “Note: There is an internal pull-up resistor” to description of pin C6
  - Added “Default: Input” and “Note: There is no internal pull-up/pull-down resistor” to description of pin D2
  - Added “Default: Input” and “Note: There is no internal pull-up/pull-down resistor” to description of pin E1
  - Added “Default: Input” and “Note: There is no internal pull-up/pull-down resistor” to description of pin E2
  - Added “Default: Input” and “Note: There is no internal pull-up/pull-down resistor” to description of pin E3
  - Corrected pin type of pin E4 from Power to Ground
  - Added “Default: Input” and “Note: There is no internal pull-up/pull-down resistor” to description of pin E5
  - Corrected pin type of pin E6 from Power to Ground

# DESCRIPTION OF CHANGES (CONTINUED)

- In Table 1 on page 8, made the following changes/corrections:
  - Corrected pin type of pin F2 from Analog to Power and changed description to “Sensor digital power (Core)”
  - Added “Default: Input” and “Note: There is no internal pull-up/pull-down resistor” to description of pins F3, F4, and F5
  - Corrected pin type of pin F6 from Analog to Power and changed description to be the same as pin F2
  - Corrected pin type of pin G2 from Power to Ground
  - Added “Default: Input” and “Note: There is no internal pull-up/pull-down resistor” to description of pins G3, G4, G5, and G6

# REVISION CHANGE LIST

**Document Title:** OV2640 Datasheet **Version:** 1.4

# DESCRIPTION OF CHANGES

The following changes were made to version 1.3:

- In Table 6 on page 11, made the following changes:
  - Added “40 mA” for Maximum specification of $I_{DDA-A}$
  - Added “35 mA (YUV)” and “50 mA (Compressed)” for Maximum specification of $I_{DDA-D}$
  - Added “10 mA” for Maximum specification of $I_{DDA-IO}$
  - Added “2 mA” for Maximum specification of $I_{DDS-SCCB}$
  - Added “1200 µA” for Maximum specification of $I_{DDS-PWDN}$

# REVISION CHANGE LIST

**Document Title:** OV2640 Datasheet **Version:** 1.5

# DESCRIPTION OF CHANGES

The following changes were made to version 1.4:

- Under Register Set section on page 18, changed the second paragraph from
  "There are two different sets for register address from 0x00 to 0x7E. Both register 0xF8 and register 0xFF control which set is accessible. When 0xF8=00 and 0xFF=00, Table 12 is effective. When 0xF8=01, 0xFF=7F, Table 13 is effective."
  to
  "There are two different sets of register banks. Register 0xFF controls which set is accessible. When register 0xFF=00, Table 12 is effective. When register 0xFF=01, Table 13 is effective."

# REVISION CHANGE LIST

**Document Title:** OV2640 Datasheet **Version:** 1.6

# DESCRIPTION OF CHANGES

The following changes were made to version 1.5:

- In Table 12 on page 18, changed name, default, R/W, and description of register 0x44 from “RSVD”, “XX”, “–”, and “Reserved” to “Qs”, “0C”, “RW”, and “Quantization Scale Factor”
- In Table 12 on page 21, changed description of register RA_DLMT (0xFF) from:

  Sensor/Device Register Address Delimiter
  <(value of register 0xFF): Sensor address
  (value of register 0xFF): DSP address

  to:

  Register Bank Select
  Bit[7:1]: Reserved
  Bit[0]: Register bank select
  0: DSP address
  1: Sensor address
- In Table 13 on page 22, changed default value for register REG08 (0x08) from “00” to “40”
- In Table 13 on page 22, changed description of register bits COM2[1:0] (0x09) from:

  00: Weakest
  01: Double capability
  10: Double capability
  11: Triple drive capability

  to:

  00: 1x capability
  01: 3x capability
  10: 2x capability
  11: 4x capability
- In Table 13 on page 22, changed default value for register PIDL (0x0B) from “40” to “41”
- In Table 13 on page 23, changed description of register bit CLKRC[6] (0x11) to “Reserved”
- In Table 13 on page 25, added “(if Bypass DSP is selected)” to description of register COM10 (0x15)

# REVISION CHANGE LIST

**Document Title:** OV2640 Datasheet **Version:** 1.7

# DESCRIPTION OF CHANGES

The following changes were made to version 1.6:

- In Table 5 on page 11, deleted row for ESD Rating, Human Body model specification
- In Figure 18 on page 16, deleted callouts for $t_{dfvf}$, $t_{dvsc}$, and $t_{dvh}$
- In Figure 18 on page 16, shortened first VSYNC cycle to match the second VSYNC cycle and added callout "$4xt_{LINE}$" to both cycles
- In Figure 19 on page 16, deleted callouts for $t_{dfvf}$, $t_{dvsc}$, and $t_{dvh}$
- In Figure 19 on page 16, shortened first VSYNC cycle to match the second VSYNC cycle and added callout "$4xt_{LINE}$" to both cycles
- In Table 11 on page 15, deleted rows for tdfvf, tdvsc, and tdvh

# REVISION CHANGE LIST

**Document Title:** OV2640 Datasheet **Version:** 1.8

# DESCRIPTION OF CHANGES

The following changes were made to version 1.7:

- Under Key Specifications on page 1, changed Power Supply (Core) specification from "1.2VDC ± 5%" to "1.3VDC ± 5%"
- Under Key Specifications on page 1, deleted Operation Temperature Range specification
- In Table 6 on page 11, changed Min, Typ, and Max for Supply Voltage ($V_{DD-D}$) from "1.14", "1.2", and "1.26" to "1.24", "1.3", and "1.36", respectively
- In Table 6 on page 11, changed Typ for Active (Operating) Current ($t_{DDA-D}$) from "25 (YUV) 35 (Compressed)" to "30 (YUV) 45 (Compressed)"
- In Table 6 on page 11, changed Max for Active (Operating) Current ($t_{DDA-D}$) from "35 (YUV) 50 (Compressed)" to "40 (YUV) 60 (Compressed)"
- In Table 6 on page 11, changed Max for Active (Operating) Current ($t_{DDA-IO}$) from "10" to "15"
- In Table 6 on page 11, added footnote a (and changed previous footnote a and b to b and c, respectively) "If using internal regulator for DVDD, $V_{DD-A}$requires greater than or equal to 2.65V" to Minimum value of Supply voltage ($V_{DD-A}$) specification
- In Table 6 on page 11, changed footnote c (previously footnote b) from "$V_{DD-A}$=2.8V, $V_{DD-D}$=1.2V, and $V_{DD-IO}$=1.8V for 15 fps in UXGA mode" to "$V_{DD-A}$=2.8V, $V_{DD-D}$=1.3V, and $V_{DD-IO}$=1.8V for 15 fps in UXGA mode"